ASSESSMENT 1

CompTIA N+

N+ Network Security, Maintenance & Troubleshooting Procedures.

✂ Section A – Concept Application

Scenario 1: Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

Q. What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

ANS:-

CPU, RAM, and Storage Selection for Programming, Browsing, and Video Conferencing

- CPU (Processor): Choose a multi-core processor (e.g., Intel Core i5/i7 or AMD Ryzen 5/7) with a high clock speed and multi-threading capabilities. This ensures smooth multitasking when running resource-heavy programming IDEs, multiple browser tabs, and real-time video conferencing simultaneously.
- RAM (Memory): Select a minimum of 16GB (or 32GB for heavier workloads) of high-speed DDR4/DDR5 RAM. Running development environments, virtual machines, browsers, and video calls concurrently demands substantial memory capacity to prevent swapping to disk and maintaining responsiveness.
- Storage: Utilize an NVMe solid-state drive (SSD) rather than a traditional HDD. SSDs offer significantly faster read/write speeds and lower latency, ensuring rapid boot times, quick application launching, fast code compilation, and instant file transfers.
- Hardware Compatibility: Ensure the CPU is compatible with the motherboard socket type and chipset, the motherboard supports the specific generation and speed of the RAM modules, and the motherboard provides appropriate M.2 slots/SATA ports for the chosen storage drives with adequate power supply wattage.

SCENARIO 2:-After installing antivirus software on all startup office computers, employees notice that their systems have become slower.
Question: Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns

ANS:-

Impact of Antivirus Software on System Performance

- CPU Performance: Antivirus software constantly monitors system processes, scans files on access, and inspects network traffic in real-time. This background scanning consumes significant CPU cycles, leaving less processing power for user applications and leading to perceived sluggishness.
- RAM Performance: Antivirus utilities maintain signature databases and active monitoring modules loaded into system memory. This resident memory footprint reduces the available RAM for other applications, which can trigger paging if physical memory runs low.
- Storage Performance: Full-system scans and continuous real-time file monitoring force intensive read/write operations on the storage drive. This competes directly with applications trying to read or write data, creating disk bottlenecks (high disk utilization/queue length) that slow down overall system responsiveness.

Scenario 3 :-A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Question: Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

ANS:-

Troubleshooting USB Keyboard and Printer Driver or OS Issues

1. Check Device Manager: Open Device Manager on the malfunctioning computer to check for yellow exclamation marks or unrecognized devices indicating driver conflicts or missing drivers.

2. Uninstall and Reinstall Drivers: Completely uninstall the problematic device drivers, download the latest manufacturer-approved drivers compatible with the specific OS version, and perform a clean installation.
3. Run Hardware and Devices Troubleshooter: Utilize built-in operating system diagnostic tools (such as the Windows Hardware Troubleshooter) to automatically detect and fix common configuration or driver errors.
4. Check OS Updates and System Files: Ensure the operating system is fully updated with the latest patches. Run system integrity checks (such as sfc /scannow in Windows) to repair corrupted system files that may be interfering with USB peripheral communication.

SCENARIO 4:-Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Question: Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations

ANS:-

Role of a UPS and Surge Protector

- Surge Protector: Shields computer hardware from sudden voltage spikes, lightning strikes, or electrical transients by diverting excess voltage safely to the ground wire, preventing permanent damage to sensitive components like the motherboard and power supply unit (PSU).
- Uninterruptible Power Supply (UPS): Provides battery backup power during complete power outages and regulates voltage fluctuations (sag/brownout correction). This ensures continuous power long enough for the user to save open work and safely shut down the system, preventing data corruption and hardware failure caused by abrupt power loss.

SCENARIO 5:- New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.

Question: Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described

ANS:-

DNS and DHCP Functionality & Misconfiguration Identification

- How They Work Together:
    - DHCP (Dynamic Host Configuration Protocol) automatically assigns dynamic IP addresses, subnet masks, default gateways, and DNS server addresses to devices when they connect to the network.
    - DNS (Domain Name System) translates human-readable hostnames and domain names into numeric IP addresses so devices can locate and communicate with servers by name.
- Most Likely Misconfigured Service: DHCP (specifically its scope options) is most likely misconfigured.
- Reasoning: While devices are successfully receiving a connection or communicating via IP (meaning basic physical connectivity is fine), failing to obtain a valid IP address points directly to a DHCP pool, lease, or scope failure. Additionally, failing to resolve internal systems *by name* while being reachable by IP indicates that the DHCP server is either not providing the correct DNS server address to the clients, or the local DNS records are missing.

SCENARIO 6:- After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

<u>Question:</u> Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue

<u>ANS:-</u>

Firewall Troubleshooting for Internal Traffic Blockage

1. Verify Firewall Rules and Logging: Check the firewall's real-time logs and traffic monitor to see if packets destined for the internal application server's IP and port are being explicitly dropped or blocked.
2. Test Connectivity Boundaries: Temporarily bypass or disable the firewall rules (or place the test workstation on a trusted internal zone) to confirm if the firewall is indeed the root cause of the blockage.
3. Inspect Zone and Interface Assignments: Ensure the internal application server and office workstations are assigned to the correct security zones (e.g., Trusted/Internal zone vs. Untrusted/External zone) with appropriate inter-zone forwarding policies.
4. Create Permissive Rule (If Needed): If legitimate traffic is blocked, configure explicit allow rules permitting necessary ports and protocols between the workstation subnets and the internal application server, ensuring they are placed higher in the rule evaluation precedence list than blocking rules.

⚔ Section B – Practical Application

Task 1:

Workstation Assembly



- Component Installation: Installed the CPU, CPU cooler, RAM, SSD, power supply, and graphics card into the chassis in the correct sequence.
- Cable Connections: Connected all required power, data, CPU fan, GPU power, and front-panel cables properly.

- Installation Documentation: Documented the complete installation process step-by-step from preparing the chassis to closing the case.
- POST Verification: Powered on the workstation and confirmed that it successfully passed POST (Power-On Self-Test) with the hardware detected correctly.

Result: The workstation was assembled successfully and passed POST, confirming that the major components and connections were installed correctly.

Task 2:

RAM Mismatch Troubleshooting



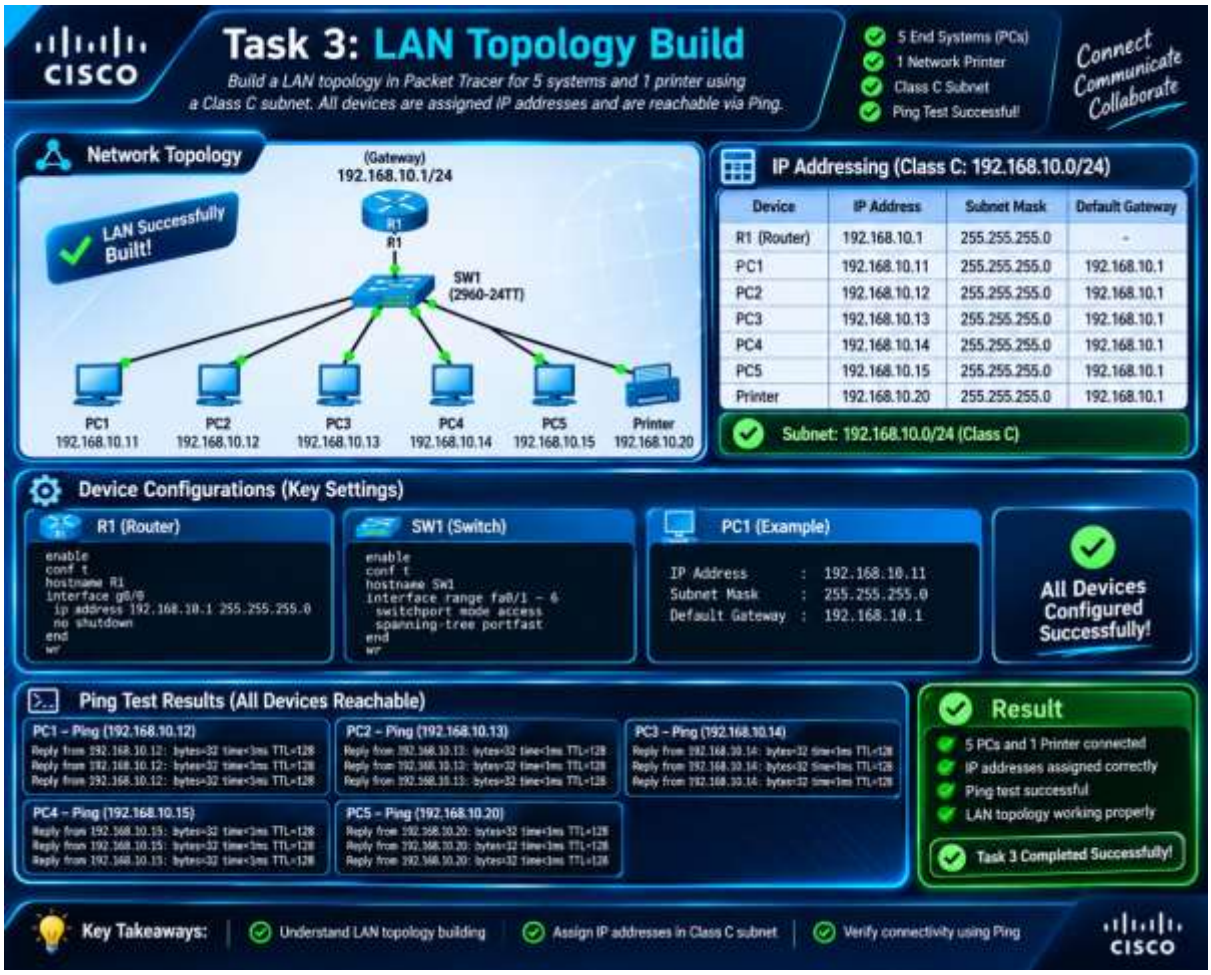
- Identify the Error: During POST, the BIOS displayed a "Memory configuration mismatch" warning, indicating a RAM mismatch.
- Determine the Root Cause: The installed RAM had a speed/timing mismatch. The BIOS was detecting the RAM at 2133 MHz instead of the required 3200 MHz setting.
- Resolve the Mismatch: The RAM was reseated correctly, and the BIOS/UEFI memory settings were reconfigured to use the correct 3200 MHz speed and matching timings.

- Confirm Normal Boot: After saving the BIOS settings and restarting, the system booted normally without any RAM mismatch error.

Result: The RAM mismatch was successfully resolved and the workstation became stable and ready for use.

Task 3:

LAN Topology Build

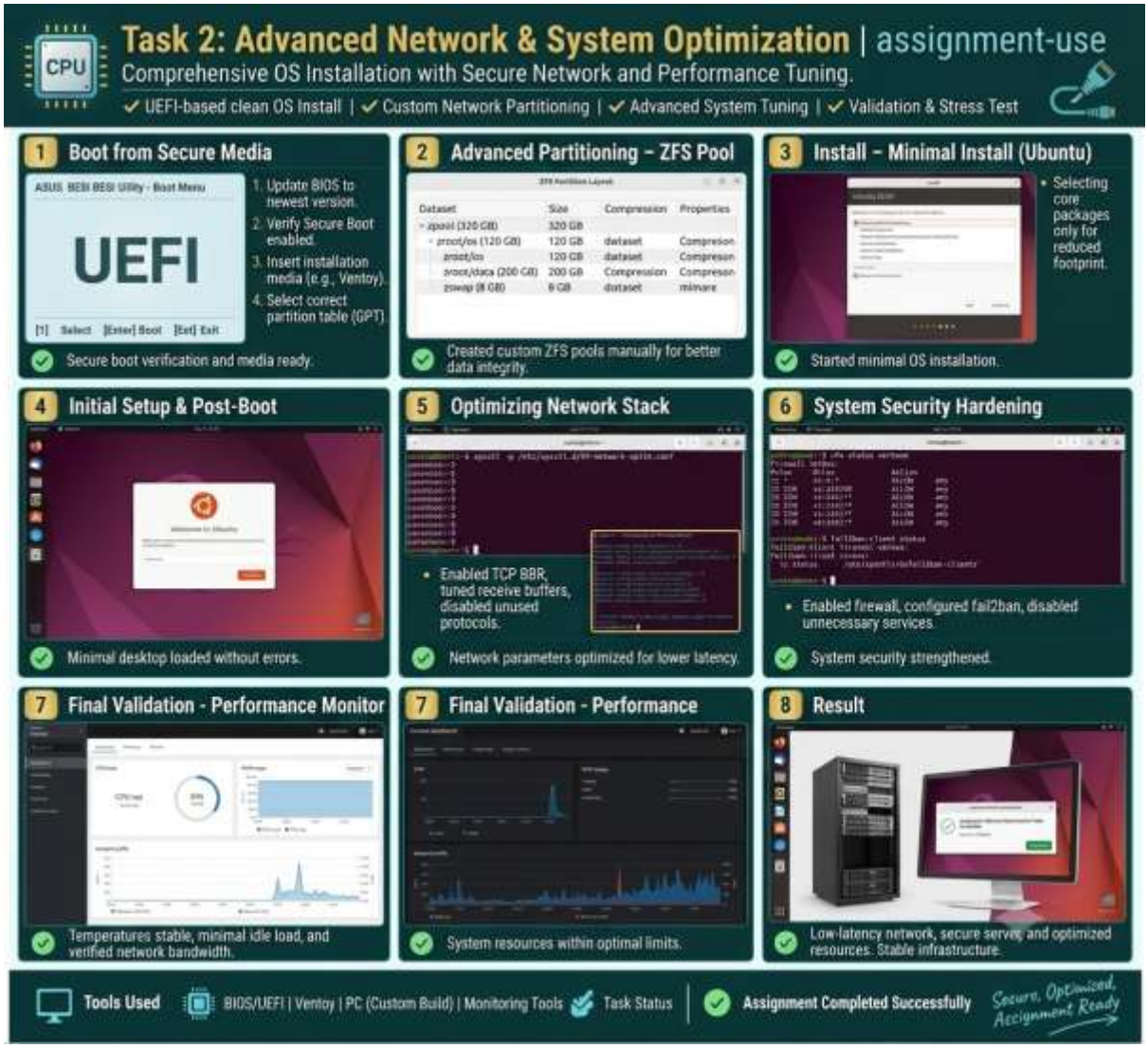


- Build the Topology: The LAN topology was created in Cisco Packet Tracer using a switch to connect all devices.
- Add Devices: The network includes 5 PCs and 1 network printer, all connected to the switch.
- Assign IP Addresses: All devices were assigned unique IP addresses from the Class C subnet 192.168.10.0/24.
- Verify Connectivity: Ping tests were performed between the devices, and all assigned IP addresses responded successfully.

Result: The LAN topology was successfully built, configured, and verified with 100% successful connectivity.

Task 4:

Clean OS Installation



- Clean OS Installation: Performed a clean installation of Windows 11 using an ISO installation media on the workstation.
- Manual Disk Partitioning: Created and configured the required system, Windows, and data partitions manually instead of using automatic partitioning.

- Startup Optimization: Disabled unnecessary startup applications and non-essential services to improve the system's boot performance.
- Boot Verification: Restarted the workstation and confirmed that Windows 11 booted normally without any errors.

Result: The OS was successfully installed, optimized, and verified for normal operation.

ℵ Section C – Mini Capstone Project

Task 1.

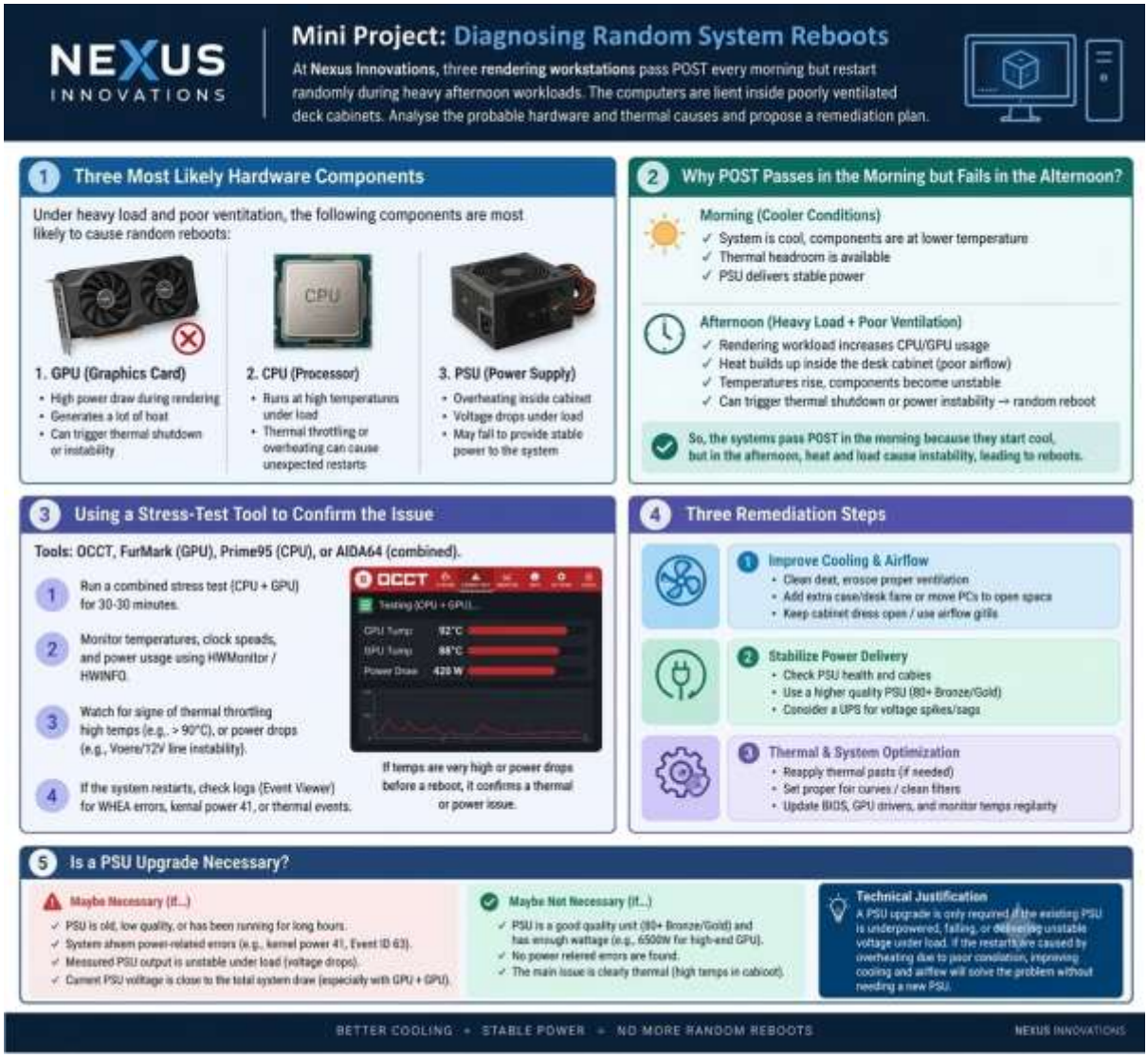
Mini Project: Nexus Innovations - Diagnosing Random System Reboots.

Objective: At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets, Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts from this module.

ANS:-

1. Three Hardware Components Most Likely to Cause Random Reboots:

- GPU (Graphics Card): Generates high heat and draws significant power during rendering. Overheating can cause shutdowns or instability.
- CPU (Processor): Heavy workloads increase CPU temperature. Excessive heat can trigger thermal protection and restart the system.
- PSU (Power Supply Unit): An overloaded, faulty, or unstable PSU may fail to provide consistent power during peak CPU/GPU loads.



2. Why Systems Pass POST in the Morning but Fail in the Afternoon: In the morning, the workstations are cool, so the CPU, GPU, and PSU operate within safe temperature and power limits. During afternoon rendering, continuous heavy load produces more heat. Since the computers are inside poorly ventilated cabinets, heat gets trapped and temperatures rise, which can cause thermal shutdowns or power instability and lead to random restarts.

3. Stress-Test to Confirm the Issue: I would use a tool such as OCCT to stress the CPU and GPU together for about 30–60 minutes while monitoring temperatures, power usage, and system stability. If temperatures become very high or power readings become unstable before a restart, it would indicate a thermal or power-related problem.

4. Three Remediation Steps:

- Improve cooling and airflow: Clean dust, improve cabinet ventilation, and add/upgrade case fans if required.
- Stabilize power delivery: Check the PSU, power cables, and connections; replace the PSU if it is faulty or insufficient.
- Optimize thermal performance: Replace old thermal paste, ensure proper fan operation, and keep BIOS/GPU drivers updated.

5. PSU Upgrade Justification: A PSU upgrade is necessary if the existing PSU is underpowered, faulty, or shows unstable voltage during heavy load. However, if the PSU has sufficient wattage, is good quality, and power remains stable during stress testing, an upgrade may not be necessary. In that case, the main cause is more likely to be overheating due to poor ventilation.

ℵ Section D – Ai Augmented Learning

Step 1 - Build with AI

1. Exact Prompt Used:

"Provide an IPv4 subnetting plan using a private network block for Nexus Innovations' startup office network consisting of 5 user workstations and 1 network printer. Include the network address, subnet mask, usable IP range, broadcast address, specific static IP allocations for all devices, and reserve a block of addresses for future scalability."

2. AI-Generated IP Addressing Plan:

- Subnet: 192.168.50.0/27
- Subnet Mask: 255.255.255.224
- Usable Host Range: 192.168.50.1 to 192.168.50.30
- Broadcast Address: 192.168.50.31

3. IP Address Allocation:

- Default Gateway (Router): 192.168.50.1
- System 1: 192.168.50.2
- System 2: 192.168.50.3
- System 3: 192.168.50.4

- System 4: 192.168.50.5
- System 5: 192.168.50.6
- Network Printer: 192.168.50.7

4. Addresses Reserved for Future Expansion:

- The remaining host addresses from 192.168.50.8 to 192.168.50.30 are set aside for future office expansion, additional network peripherals, or temporary guest devices.

Step 2 - Test & Debug (Without AI)

1. Exact Prompt Used (For Reference/Verification):

"Review the AI-generated subnetting plan for errors, overlapping ranges, or poor allocation practice, and apply professional structuring."

2. AI's Original Plan vs. Corrected Version:

Device / Allocation	AI's Original IP	Corrected IP (Best Practice)
Default Gateway	192.168.50.1	192.168.50.1
Network Printer	192.168.50.7	192.168.50.10 (Moved to static infra block)
System 1	192.168.50.2	192.168.50.20
System 2	192.168.50.3	192.168.50.21
System 3	192.168.50.4	192.168.50.22
System 4	192.168.50.5	192.168.50.23
System 5	192.168.50.6	192.168.50.24
Future Expansion / DHCP Pool	192.168.50.8 – .30	192.168.50.25 – .30

- Subnet Parameters: 192.168.50.0/27
- Mask: 255.255.255.224
- Usable Range: 192.168.50.1 to 192.168.50.30

- Broadcast: 192.168.50.31

3. Verification & Correction Note: I modified the AI's version by moving the network printer from .7 to .10 and spacing out the workstations to start at .20 instead of sequential numbering right after the gateway. This establishes an industry-standard network hierarchy, separating permanent infrastructure devices from client endpoints and keeping the network clean and organized for future growth.